DeepMind's WeatherNext Adds a Day of Cyclone Warning and Opens Its Weights

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2026-09-15

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Google DeepMind's WeatherNext Cyclones predicts a tropical cyclone's track, intensity, and wind structure about a day earlier than leading operational models. Across storms from 2023 to 2025, the team reports an average lead-time gain of at least 24 hours, an improvement it compares to a decade of conventional progress. It trains end to end on nearly 20 terabytes of atmospheric data and a database of about 5,000 historical storms. It generates 1,000-member

ensembles, up from 50, each one a plausible future, and it produces a 15-day forecast in under a minute on a TPU. DeepMind published the results in Nature and released the code and weights under Apache 2.0.

Forecasters used an earlier version in the 2025 Atlantic season, when it called Hurricane Melissa's rapid intensification and Jamaica landfall five days out, at 80% confidence and near certainty three days before. Rapid intensification, a jump of at least 35 miles per hour in 24 hours, is what makes them so dangerous. The National Hurricane Center's verification report named it the top individual model for both track and intensity that season. Open weights mean weather services and researchers can run it without a vendor contract, and a miniature version fits a free Colab notebook.

The result also challenges a long-standing assumption. WeatherNext reads coarser input than the regional models it outperforms, evidence that high resolution is not required for accurate intensity forecasts and that low-resolution data carries more storm signal than meteorologists recognized.

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Sources:

- WeatherNext: AI model achieves breakthrough in forecasting cyclones (Google DeepMind)
- Operational tropical cyclone forecasting with AI (Nature, open access)
- google-deepmind/weathernext (GitHub, Apache-2.0)
- Our WeatherNext 2 AI model demonstrated a massive leap forward in predicting cyclones (Google)
- How WeatherNext helped the NHC better predict Hurricane Melissa's landfall (Google DeepMind)

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